The Shackleton rover: An Expedition to the Lunar South Pole

Patron
The Honourable Alexandra Shackleton

Coordinator
Patrick Harkness, Professor of Exploration Technology, University of Glasgow

Contact
patrick.harkness@glasgow.ac.uk

Investment Brief

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The Steam Yacht *Endurance*, trapped in ice, 1915. Picture by Frank Hurley.

Concept

Proposal: The UK should build a small lunar rover with a drilling payload. The objective would be to make first contact with subsurface water-ice near the lunar south pole. The package would be delivered by US Commercial Lunar Payload Services.

Vision: The rover will be named *Shackleton*, with a mission inspiration value focussed on ambition, determination, and perseverance. This is anchored with the ice connection, the south polar target, and by our patron, The Hon. Alexandra Shackleton. The objective is to boost national morale, spark new careers in STEM, and to position the UK both as a custodian of the lunar environment and a participant in the nascent lunar resource economy.

Competition: Peer nations are active. Australia is building *Roo-ver*, which will not seek to penetrate the terrain. The European Space Agency is building *MAGPIE*, which will attempt to drill to 10cm. Canada recently cancelled the *Canadian Lunar Rover* and instead moved focus to a larger vehicle. The US is working on *VIPER*, which is a larger vehicle also.

Differentiation: Smaller (and therefore faster, cheaper) missions have ground penetration challenges because the weak lunar gravity limits the downforce available to the drill. The UK leads on several technologies, such as localised ultrasonic terrain fluidisation, which can address this issue. A small UK rover could achieve a greater depth than a larger competitor, on a faster timeline, increasing our chances of making first contact with the water-ice below.

Further details are available at www.shackletonrover.co.uk

Objectives

Accessing the lunar resource economy: Water-ice is a vital resource because it can be split into hydrogen and oxygen. These are excellent rocket propellants. Given that around 90% of the launch mass of any rocket is propellant for immediate use on climbout, having access to propellants already in space vastly simplifies logistics. This is an enabler for the lunar economy, which is larger than the space resources market itself. From PwC:

The size of the [space resources utilisation] market, expected to be cumulatively worth about $63 billion by 2040, [is] driven primarily by the demand for propellant.

https://www.pwc.com.au/industry/space-industry/lunar-market-assessment-2021.pdf

New careers for space: The non-incremental vision and rapid development programme will provide experience across the entire mission lifetime, creating a cohort of people able to deliver a national resurgence in space.

Achievement, ambition, and national morale: This will be a scientific achievement of the highest magnitude. It will have a clear focus, it will have a compelling narrative, and it will demonstrate that British engineering can still lead the world.

Becoming a custodian of the moon: The preservation of Antarctica is delivered by the Antarctic Treaty System. The UK has a consultative role in this system due to our established operations on the ground. As lunar regulation emerges, similar credentials may be required.

Sectoral interests

The lunar economy will surely continue to emerge. The strategic interests of the United States and China are clear, and the wider activity will always require propellants that are extremely expensive to deliver from Earth. A permanent presence on the moon will also require metals, concrete, and water, ideally created in-situ. However, there is still time for the UK and its industries to enter the sector as early adopters, shaping the regulation and building the heritage required to exploit the growth of this activity in the decades ahead.

Some sectoral interests include:

Mining and quarrying: *Shackleton* will pioneer lunar prospect targeting, linking established orbital detections of water-ice to ground truth. Ultrasonic terrain fluidisation will demonstrate novel approaches to excavation, transport and beneficiation as operations scale. Autonomous operations will develop capabilities relevant to future fleets of surface vehicles.

Earthmoving: Permanent facilities will require excavation work and established heavy plant manufacturers may wish to establish an early position. Lunar gravity will reduce load reaction and the sharp, uneroded lunar soil may resist excavation. Terrain characterisation by *Shackleton* will therefore provide vital engineering data to inform future machine design.

Construction: The manufacture of concrete and the extraction of metals from lunar soil is an active field of research, with the UK in a leading position. However, experiments remain confined to simulated soil. *Shackleton* could carry a payload to create the first construction materials on the Moon, positioning British industry to supply future landing pads and bases.

Industrial gases: Propellant production will require extraction, processing, and handling of hydrogen and oxygen. Plant design will depend on whether polar water occurs as frost, interstitial deposits, or glaciation. *Shackleton* can establish this ground truth, facilitating gas and process engineering companies seeking to enter the lunar propellant industry.

Nuclear energy: Sustained operations in permanent shadow require power independent of sunlight. The UK has growing capability in space nuclear power, including americium radioisotope systems for heating and electricity. *Shackleton* offers an early application for these technologies, connecting Britain’s established nuclear sector to the emerging market.

Telecom and navigation: Every lunar operation depends on comms and positioning. The UK has a leading role in ESA’s Moonlight programme, and *Shackleton* offers an early proving ground for British receivers, antennas and navigation systems. Flight heritage could position suppliers within the technology stack used by future fleets of lunar vehicles.

Farming and fertilizers: Fertilised lunar soil can support plant life, and agriculture is a requirement for self-sufficiency. Ploughing and cultivation raise familiar challenges in traction, power take-off, and autonomous precision operations, while the necessary closed-loop management of nitrogen and phosphorus creates a direct link to the fertilizer industry.

Insurance: Expanding lunar infrastructure and autonomous vehicle fleets will require new approaches to pricing risk. *Shackleton* will generate real reliability data through exposure to dust, radiation, thermal cycling, and terrain hazards. Learning from *Beagle2*, we will be keen to work with insurers to establish the actuarial foundations of an emerging lunar economy.

Timeline

2 October: Requirements definition meeting (Leicester)

30 October: Schedule and cost report published

18 November: Pitch meetings (London)

Core Technical Team

Coordinator: P. Harkness, University of Glasgow, leads on ultrasonic terrain penetration

Delivery to lunar surface: B. Bussey, Intuitive Machines, Chief Scientist

Architecture: S. Barber, Open University, experience of Beagle2 on Mars

Instrumentation: N. Bowles, University of Oxford, extensive flight experience on Moon

Operations: S. Banham, Imperial College, experience of Curiosity rover on Mars

Costing: J. Martin-Torres, Northumbria University, extensive flight experience on Mars

Terrain Interactions: H. Sargeant, University of Leicester, leads on in situ resources

Outline Pitch

The team estimates £25m to deliver *Shackleton*, subject to our schedule and cost report. These funds are not available from public sources. However, we believe the time is right:

- Water-ice is a key to the lunar economy. Although it has been detected from orbit, ground truth is still absent. This is vital for prospecting, extraction, and beneficiation.
- Peer-nations are seeking this ground truth. If we choose to join them, we have the personnel required to make a fast and highly competitive intervention.
- There is a window of opportunity during which our unique technologies can permit a small rover to achieve significant drill depths before larger landers start to dominate.
- The moon landing will provide a huge boost to national morale. We will showcase ambition and demonstrate that the UK is still a country where anything is possible.
- We will boost engagement by echoing the heroic age of polar exploration, reimagined for the 21st century. We will even fly a fragment of *RRS Discovery* to the Moon.
- The inspiration value will raise the social capital of science and engineering and encourage young people to consider these most productive careers in future.
- We will develop a persuasive case to extend the UK's Antarctic model and position ourselves as a leading nation in lunar regulation and environmental protection.
- The lunar economy will emerge regardless of our choices. But by engaging now, British businesses will attain the highest salience and start to build real heritage.
- This heritage is essential if we are to participate in the future lunar economy. We are unlikely to mine ice and make concrete without a *Shackleton*-like pathfinder mission.
- The *Shackleton* team would wish to work with British industry to make this happen, and we look forward to meeting sectoral leaders who share our vision.

Executive Summary

Shackleton is a proposal for the UK to build a small lunar rover with a drill payload, in pursuit of first contact with the water-ice known to exist near the lunar south pole. This will be a momentous achievement because ground truth for water on the moon is currently absent: it may be frost, ice, or entirely underground. This information shapes the manner in which the water should be exploited, and it is ultimately key to a supply of hydrogen and oxygen in space. These materials are ideal rocket propellants, and they are vital to any lunar economy.

Small prospecting rovers can now be delivered relatively quickly and cheaply. Lunar gravity, however, provides little reaction force for drilling. The UK leads on relevant technologies, including local ultrasonic terrain fluidisation, which reduce penetration resistance. A small British rover may reach greater depths than considerably larger vehicles, increasing our chances of making contact with subsurface ice in the shorter term.

This would be a scientific and industrial milestone of global importance. *Shackleton* can help us to exploit it. The lunar economy will require excavation, construction, power, navigation, and insurance. The UK has considerable capability in these sectors, but little lunar heritage. *Shackleton* would allow British companies to begin to acquire it through participation in an operational lunar mission. If we want to secure a first-mover advantage, we must start here.

Most importantly, however, we seek to inspire. *Shackleton* will boost national morale, build careers in science and engineering, and position the UK for the next century in space. We will leverage a convincing narrative of polar exploration, echoing Sir Ernest Shackleton's heroic mission by carrying a fragment of the *RRS Discovery* to the moon, just as the Ingenuity helicopter carried a fragment of the *Wright Flyer* to Mars. Our patron, The Honourable Alexandra Shackleton, will underline this spirit of ambition and determination.

With success, we will be better able to shape the future of space. The largely privately-funded *Discovery* expedition demonstrated the UK’s ability to operate independently in Antarctica 125 years ago. This positioned us as a founder nation of the Antarctic Treaty System, which preserves Antarctica as a peaceful global commons to the present day

To earn the same hearing in lunar affairs, we must similarly demonstrate a presence on the ground. Ultimately, we have a chance to do something difficult, visible, and consequential, and yet something which is still within the grasp of a country such as our own, provided that we start now. It is difficult to see when the next opportunity set with such potential impact will arise. Larger nations, with larger rovers and larger budgets, are getting ready. We can still compete, but only if we go while we can still exploit our smaller technologies.

Therefore, a requirements definition meeting in Leicester on 2 October will develop a draft requirements slate and CONOPS, examining how choices in drilling depth, location and mission duration affect the mission cost. This will establish a credible programme schedule and cost. Without prejudice to this process, our current estimate is £25m.

However, we are aware that this is outside the public budget. Therefore, a smaller London meeting on 18 November will present the proposition to potential sponsors. We intend to explore the *Discovery* model of industrial and private sponsorship, with oversight and support from national agencies. Companies with a future interest in the lunar economy can establish an early position, while a British lunar landing can attract broader national support.

We would be delighted to engage with partners interested in supporting our concept.